

HitPredict

Predicting Billboard Hits Using Spotify Data

Project 21 | UE24CS352A Machine Learning | E Section

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1. Problem and objective

Classify the original HitPredict dataset's Billboard hit label using nine Spotify audio features. The task is binary supervised classification: hit (1) or sampled non-hit (0). We evaluate generalization to artists absent from training, while treating scores as historical classification outputs rather than calibrated future-hit probabilities.

2. Dataset and preparation

We use the no-date CSV from the public archive linked in the Stanford CS229 study [1]. Its 4,040 rows reduce to **4,019 unique artist/title records** after removing 21 duplicates: 1,595 hits and 2,424 non-hits across 2,181 normalized artists. No missing audio values or conflicting duplicate labels occur. Three zero-tempo values remain as supplied.

The inputs are danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence and tempo. Names identify groups only. ArtistScore is excluded because its prior-hit timing cannot be audited in this file. Key and mode are also excluded. The dated CSV is preserved only as a reference.

Partition	Tracks	Hits	Groups
Training	2,554	1,011	1,395
Validation	641	268	349
Test	824	316	437

3. Approach and implementation

Normalized artist names and identical audio vectors stay within one group. A fixed seed (21) reserves 20% of groups for testing, then 20% of the remainder for validation. A pipeline fits median imputation and standardization inside each training fold [2]. The model is chosen by mean 3-fold group cross-validation ROC-AUC. The validation set chooses a cutoff that maximizes F1 over 0.10-0.90. Test labels do not drive either choice.

Five model families and a majority baseline span nine configurations. The best configuration is a random forest with 250 trees, maximum depth 16 and minimum leaf size 5. The table shows the best training CV result per family.

Model family	CV ROC-AUC	Fold SD
Random forest	0.801	0.008
Neural network (16)	0.787	0.021
RBF SVM	0.783	0.015
Logistic regression	0.772	0.022
Linear discriminant analysis	0.766	0.026
Majority baseline	0.500	0.000

The modular Python implementation saves the pipeline, split manifest, predictions, error examples and figures. An executed notebook explains the experiment. Streamlit and the batch tool share one scoring function. Checks cover group separation, preprocessing, saved predictions, invalid inputs and demo interactions.

4. Results and interpretation

The selected forest achieves **test ROC-AUC 0.814** on 824 tracks. A 500-resample bootstrap of held-out groups gives a 95% interval of 0.773-0.848, conditional on this fitted model and split. Average precision is 0.681 against a test hit prevalence of 0.383. The validation-selected cutoff is 0.29.

Test metric	Chosen cutoff 0.29	Fixed cutoff 0.50	Majority
Accuracy	70.87%	74.76%	61.65%
Precision	57.69%	68.24%	0.00%
Recall	90.19%	63.92%	0.00%
F1	0.704	0.660	0.000
ROC-AUC	0.814	0.814	0.500

At cutoff 0.29, the model detects 285 hits and misses 31, while incorrectly flagging 209 non-hits. This choice favors recall. The fixed 0.50 reference improves precision at the cost of missed hits; it is not a new cutoff optimized on test outcomes. Training ROC-AUC is 0.982, indicating remaining overfitting relative to the test result.

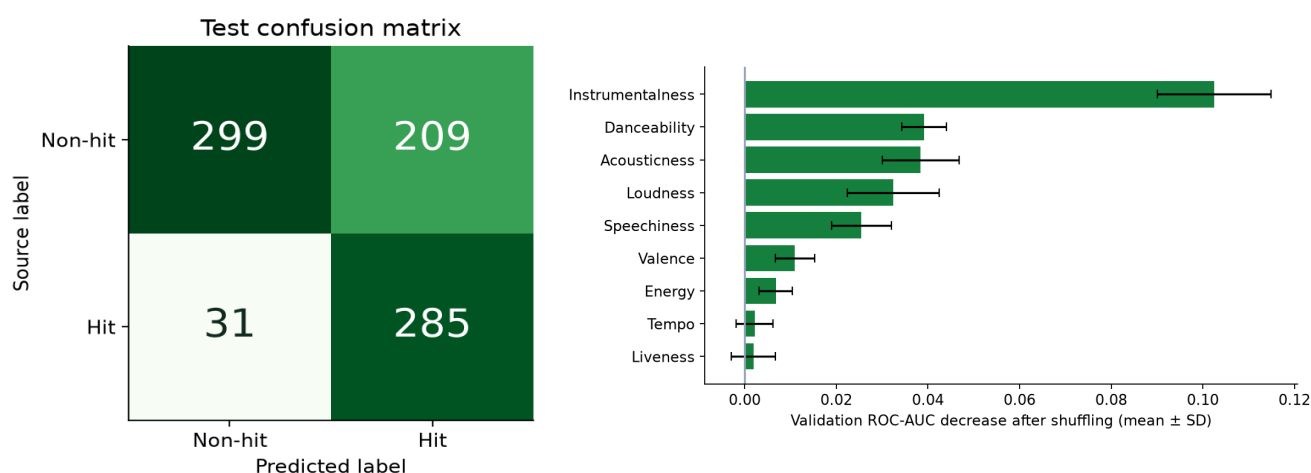


Figure 1. Left: primary test confusion matrix. Right: validation permutation importance (15 shuffles per feature; error bars show SD). Instrumentalness, danceability and acousticness have the largest average importance. These are predictive associations, not causal effects [3].

5. Limitations and conclusion

The source report gives inconsistent collection and label-window descriptions; we inherit its labels without independently rebuilding chart histories. The main file has no release dates, preventing a prospective temporal test. Artist aliases and collaborations may evade name grouping. Sampled non-hits, historical musical styles and class prevalence may differ from today's releases. Scores therefore should not be read as real-world success probabilities.

The secondary row-split diagnostic scores 0.789 ROC-AUC and shares 392 artists across splits. Different samples and training sizes prevent a causal interpretation of its difference from the grouped result. Marketing, audience and playlist exposure are unobserved.

Conclusion: Audio features provide useful signal for historical hit-label classification, but high recall comes with many false positives. A stronger next study should verify identities and release dates, define a fixed future chart window, sample representative non-hits and evaluate on later releases before assessing calibration or use in music promotion.

References and reproducibility

[1] Georgieva, E., Suta, M., and Burton, N. (2018). HitPredict, Stanford CS229. Report and public archive link: cs229.stanford.edu/proj2018/report/16.pdf. Retrieved 7 October 2026. Dataset credit retained; implementation is new.

[2] scikit-learn. Group cross-validation and leakage guidance: scikit-learn.org/stable/modules/cross_validation.html. [3] Permutation importance: scikit-learn.org/stable/modules/permutation_importance.html.

Run: `python -m src.train`; demo: `python -m streamlit run app.py`. Python 3.12.6, scikit-learn 1.9.1. Full metrics and source hashes are in the repository.